ASSIGNMENT 02 · CO3 + CO4

Reconstruct or Create?

A comparative study of four Generative AI models

Autoencoder · VAE · DCGAN · Diffusion

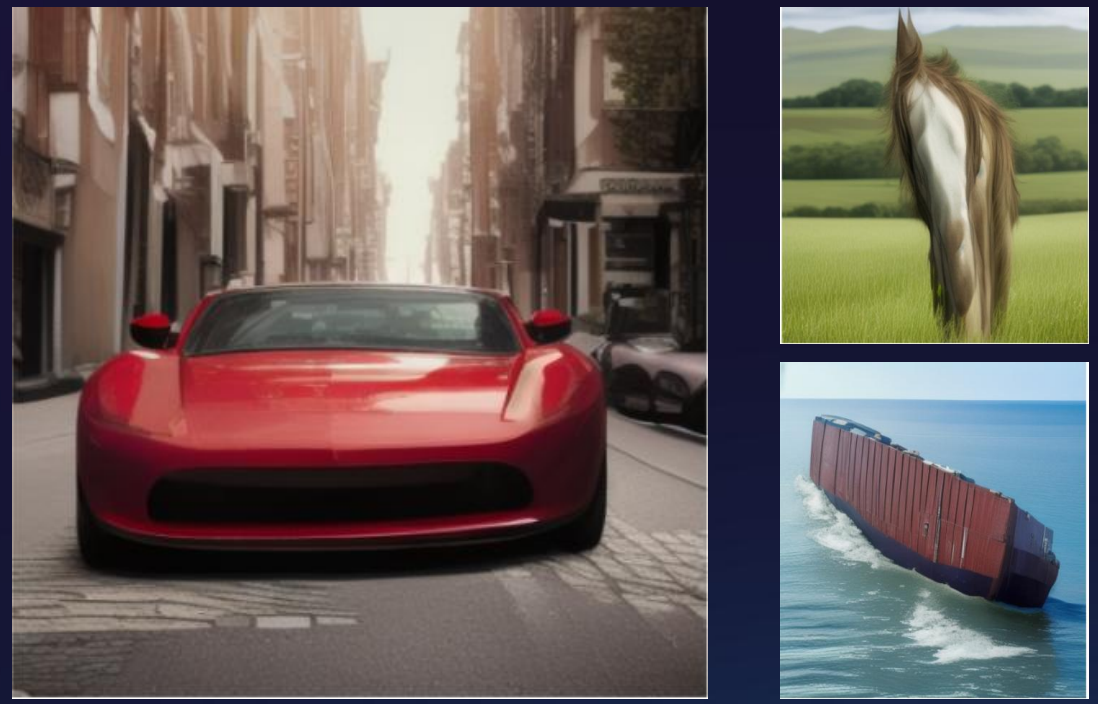
Generative AI · B.Tech AIML · MIT Academy of Engineering, Pune

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Pretrained diffusion outputs (segmind/tiny-sd), 512×512



All images shown were generated by our DCGAN (below) or the diffusion model (above).

Four models, two jobs

RECONSTRUCTION · CO3

GENERATION · CO4

Autoencoder

Squeezes the image into 128 numbers, then rebuilds it (MSE loss)

- Simple, fast, stable
- Blurry, loses detail

Used for
Compression, denoising

VAE

Learns a distribution (μ, σ), samples z = μ + σ·ε (recon + KL loss)

- Can sample new images
- Even softer outputs

Used for
Representation learning

DCGAN

Generator fakes images from noise; discriminator judges real vs fake

- New images from our data
- Unstable, artefacts

Used for
Image synthesis, augmentation

Diffusion

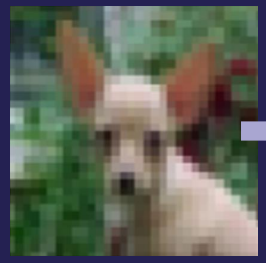
Removes noise step by step, guided by a text prompt (U-Net)

- Best quality + prompt control
- Heavy compute

Used for
Text-to-image, design

Inside each model

Autoencoder compress → rebuild



Encoder
Conv
32→64→128

z
128-d

Decoder
ConvT
64→32→16

Input image → latent code → reconstruction (see slide 4)

VAE learn a distribution, then sample

Encoder
Conv layers

μ

σ

sample
z = μ + σ·ε

Decoder
ConvT

KL term keeps the latent space close to N(0, I)

DCGAN generator vs discriminator

+ real CIFAR-10 images

noise z
100-d

Generator
Dense + ConvT
+ BatchNorm

fake
32×32

Discriminator
Conv + Dropout

Real? / Fake? — trained in turn (adversarial BCE loss)

Diffusion denoise step by step, guided by text

prompt: "a realistic photograph of a red sports car…" · guidance 7.0

pure noise
(latent)

U-Net removes noise
× 20 steps
(text-conditioned)

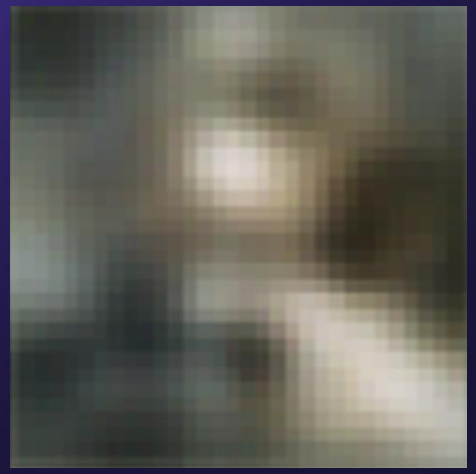


A VAE decoder turns the final latent into the 512×512 image

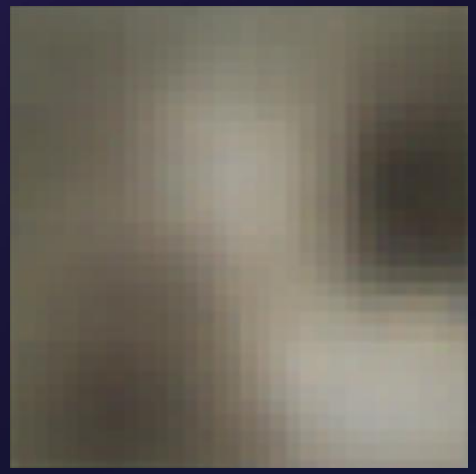
Activity A: AE keeps more detail than VAE



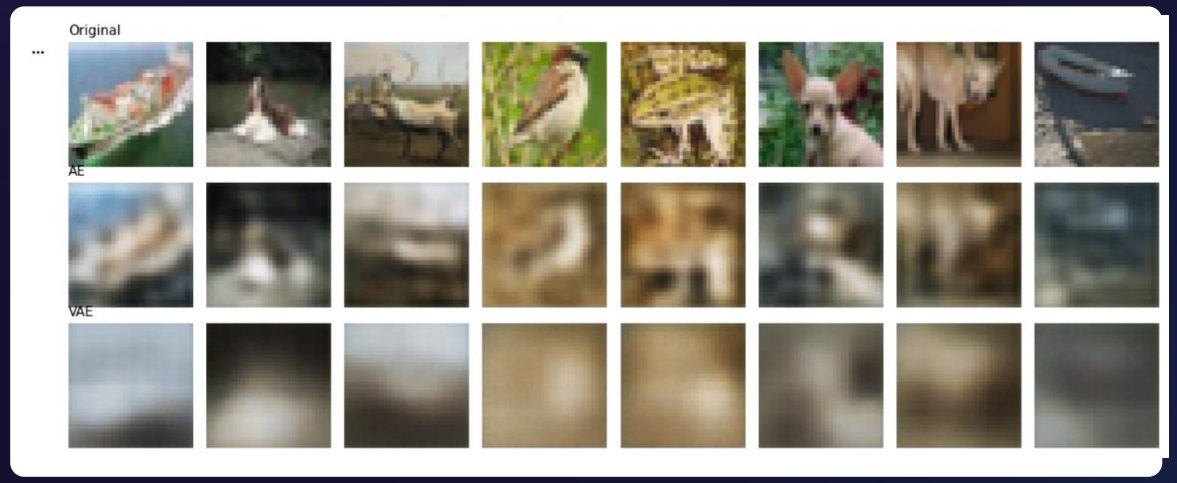
Original



Autoencoder



VAE



Notebook output · top row original, middle AE, bottom VAE (8 validation images)

0.0086

AE validation MSE

−68%

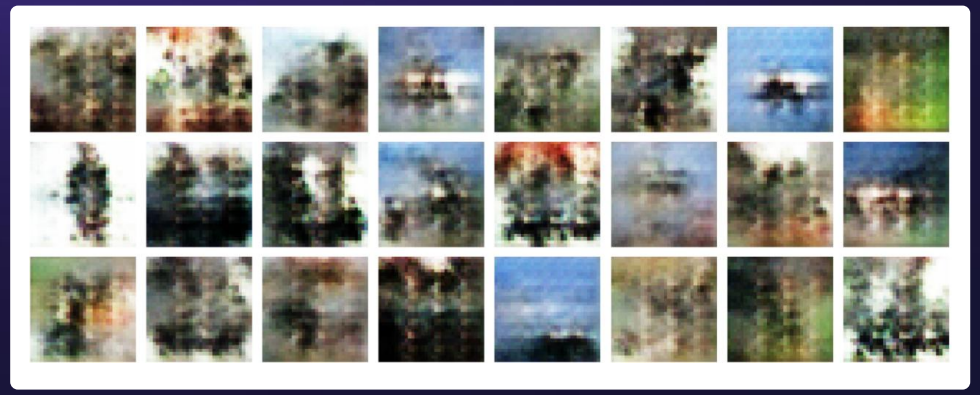
MSE drop in 15 epochs

Validation set (1,000 images)

AE PSNR 21.15 dB · SSIM 0.650
VAE PSNR 17.05 dB · SSIM 0.343

- AE keeps colour, layout and rough shape
- Fine texture and edges get blurred
- VAE images look smoother and “averaged”
- Main difference: loss of detail, not distortion

Activity B: diffusion looks far more realistic



VS



DCGAN · TRAINED BY US

40 epochs · 32×32 · final G 2.84 / D 0.61

DIFFUSION · PRETRAINED

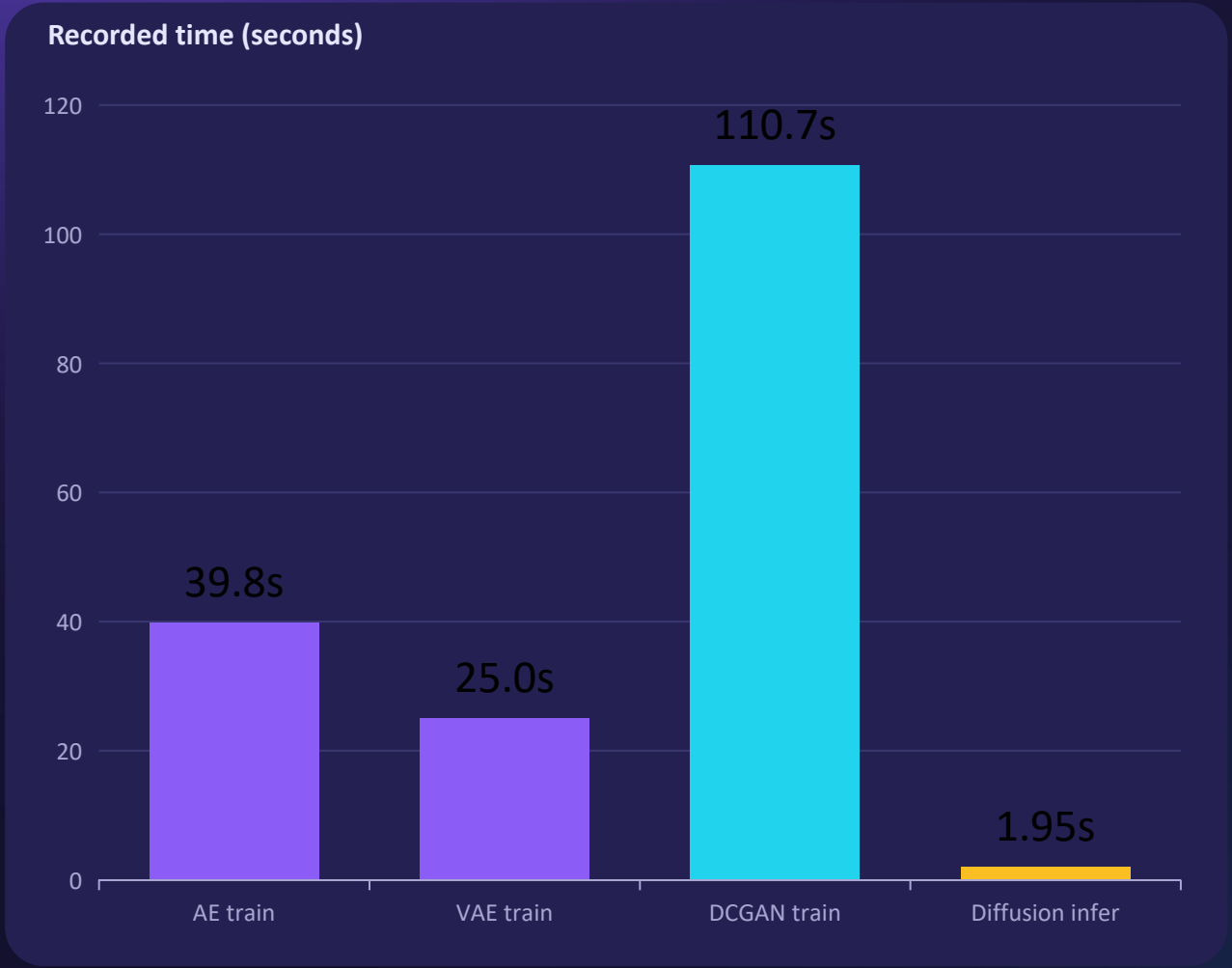
segmind/tiny-sd · 20 steps · guidance 7.0 · 1.95 s

Prompt: "A realistic photograph of a red sports car parked on a street, soft natural daylight, high detail."

What we saw

- Generator makes new images from random noise
- Samples are colour blobs; no clear objects
- D loss stayed below G loss (≈0.6 vs ≈2.8)
- Quality judged visually, not by loss
- Diffusion: sharp 512×512, a few distortions (horse, ship)

DCGAN costs the most; losses are not comparable



2.8–4.4×

DCGAN training time vs AE / VAE

2 nets

trained every step

Model	Loss and scale	Final
AE	MSE, averaged per pixel	0.0086
VAE	BCE + KL, summed	631.2
DCGAN	Adversarial BCE	G 2.84 / D 0.61
Diffusion	Pretrained	—

Different loss functions, different scales: compare the images, not the numbers.

What we learned and how we used AI

AI usage and verification

- Claude for concepts, code explanation and the report
- Every number checked against the notebook output
- Theory cross-checked with original papers and docs

Key learning

- AE vs VAE: reconstruction detail vs organised latent space
- GAN: loss values alone do not indicate image quality
- Diffusion: high-quality, prompt-guided generation
- Training time and inference time are different

Contributions

- Ashis: AE and VAE experiments
- Sarthak: DCGAN and diffusion experiments
- Pratik: comparison table, AI documentation, report and PPT

Conclusion AE/VAE for reconstruction · DCGAN for dataset-specific generation · Diffusion for the best quality at the highest cost